Course Title	Signals and Systems	**Course Code**	EE2008
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	**Credit Hrs.**	3+1
Knowledge Area	Telecommunication (KA05)	**Grading Scheme**	Relative
HEC Knowledge Area	Engineering Foundation	**Applicable From**	Spring 2026
SDG	4 | Quality Education	**PBL**	1
Pre-requisite(s)	MT2003		

Course Objective	To equip students with a solid foundation in the analysis and processing of continuous and discrete-time signals and systems, through time-domain and frequency-domain techniques.

No.	Assigned Program Learning Outcome (PLO)
1	Apply knowledge of mathematics, natural science, engineering fundamentals and Engineering specialization to the solution of complex engineering problems.
2	Identify, formulate, conduct research literature, and analyze complex Engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences

A = Assignment, Q = Quiz, M = Midterm, F=Final

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Apply fundamental concepts of signals and systems to classify and represent various types of signals and systems.	Q1, M1, A1	C3	1
2	Analyze continuous and discrete-time Linear Time-Invariant (LTI) systems using convolution operations.	M1, F	C4	2
3	Apply Fourier series and Fourier transform representation techniques to signals encountered in engineering domain.	A2, Q2, M2	C3	1
4	Analyze discrete time signals and systems using various frequency domain techniques including z-transform and DFT.	Q3, A3, F	C4	2

Text Books	**Title**	Linear Systems & Signals, 2nd Edition
	Author	B.P. Lathi
	Publisher	Oxford Pakistan
Reference Books	**Title**	Signals & Systems
	Author	Oppenheim, Alan V., Alan S. Willsky and S. Hamid Nawab
	Publisher	Pearson Education

Week	Course Contents/Topics	Chapter*	CLO*
1	Signal, Size of a Signal, Operations, Classifications, and Some useful Signal Models	1	1
2	Systems, Classification, Some System Models	1	1
3	System response to External Input: The Convolution Integral	2	2
4	System response to External Input: Graphical Convolution Operation	2	2
5	Trigonometric Fourier Series	6	3
6	Exponential Fourier Series	6	3
7	The Fourier Transform for Aperiodic Signals (Physical Appreciation)	7	3
8	Fourier Transform of some useful functions	7	3
9	Properties of Fourier Transform	7	3
10	Sampling: Bridge from Continuous Time to Discrete Time	8	3
11	Discrete-Time Signals and Systems	3	4
12	The z-transform, its application to some common signals	5	4
13	Properties of z-transform, its application to solution of Difference Equation	5	4
14	Discrete Fourier series (DTFS), Discrete-Time Fourier Transform (DTFT)	9	4
15	Discrete Fourier Transform (DFT), Fast-Fourier Transform (FFT)	8, 9	4

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterm (I+II)	30.0%
Final Exam	50.0%

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